Paper 13.25 - Toolkit for Quark-Face Transport

CQE-paper-13.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-13.25.md

Paper 13.25 - Toolkit for Quark-Face Transport

Purpose

This support paper exposes the tools for Paper 13. It is not the proof-carrying paper. The proof is the finite algebraic closure in Paper 13; this toolkit shows how to inspect the same closure digitally and by ordinary visible marks.

Digital Toolkit

Run:

```
python production/formal-papers/CQE-paper-13/verify_quark_face_transport.py
```

The verifier checks:

```
shell-2 chart state count
trace-2 J_3(O) idempotent receipt
S3 closure over the trace-2 triple
exact n=3 SU(3) group-ring closure
bounded G2/F4/T5A classifier honesty boundary
six-face color/anticolor analog consistency
```

The receipt is:

```
production/formal-papers/CQE-paper-13/quark_face_transport_receipt.json
```

Minimal Analog Surface

Use three tokens labelled:

```
C-  C0  C+
```

Write the associated chart states:

```
C- = 110
C0 = 101
C+ = 011
```

Write the associated idempotents:

```
C- = E11 + E22
C0 = E11 + E33
C+ = E22 + E33
```

Now apply each of the six permutations of `(1,2,3)` to the idempotent labels. Every result must return to one of the three labels. That is the analog check for S3 closure.

Six-Face Exposure

The optional color/anticolor exposure layer uses:

```
R, G, B, anti-R, anti-G, anti-B
```

with:

```
R -> G -> B -> R
anti-R -> R
anti-G -> G
anti-B -> B
```

This is a workbook visualization of color-face transport. It is not the physical Standard Model proof.

Hidden-Guess Diagnostic

Before revealing the receipt, ask the reviewer or model to choose:

```
Does S3 close on the trace-2 triple?
Does exact n=3 closure have zero residual?
Does the bounded route derive the bit from depth alone?
Does the color/anticolor model prove physical quark color?
```

Only after the guess is recorded reveal the receipt:

```
yes
yes
no
no
```

The diagnostic trains the boundary that matters most in this paper: algebraic closure is closed; physical derivation is not silently promoted.

Boundary

This toolkit may reproduce the Paper 13 algebraic transport receipt. It may not convert the color analogy into a physical derivation without a separate claim contract and calibrated evidence.